

Breathiness Preservation Benchmark

Objective evaluation of TTS model breathiness retention from reference audio

Date	2026-05-24
GPU	NVIDIA RTX 3060 Laptop (6GB VRAM)
Python	3.12 (Chatterbox, Kokoro), 3.11 (XTTS-v2)
Reference dataset	VCTK Corpus 0.92 (3 female pairs, Southern England + mixed accents)
Models	3 (Chatterbox 0.1.7, Kokoro 0.9.4, XTTS-v2 0.22.0)
Total audio	24 WAV files (6 references + 18 generated outputs)
Primary metric	Praat CPP (Cepstral Peak Prominence) over voiced intervals
Supporting metrics	HNR (Harmonic-to-Noise Ratio), Spectral tilt

This benchmark measures whether voice-cloning TTS models preserve breathiness cues present in reference audio. The gate mechanism validates that reference clips separate breathy from neutral before model rankings are accepted.

Methodology

1. Reference Selection

All 62 female VCTK speakers were processed for sentences 001-010. CPP and HNR were extracted per speaker (best sentence by voiced ratio). Speakers ranked by combined CPP+HNR z-score. Bottom 3 = breathiest, top 3 = most modal. Paired by matching sentence.

Pair	Sentence	Breathy speaker	CPP	HNR	Modal speaker	CPP	HNR	Region note
001	s002	p240	10.52	18.08	p229	13.36	21.41	Southern England
002	s005	p253	11.14	20.49	p301	15.65	22.12	Welsh vs American
003	s006	p264	9.52	20.37	p282	14.55	21.48	Scottish vs English NE

All paired by same sentence within pair. Both CPP and HNR direction verified before pairing.

2. Audio Generation

Each model generated 6 outputs (3 pairs × 2 conditions) using reference voice cloning where supported:

Model	Cloning	Parameters	Sample rate
Chatterbox 0.1.7	Yes (audio_prompt_path)	exaggeration=0.5	24 kHz
Kokoro 0.9.4	No (default voice)	voice=af_bella	24 kHz
XTTS-v2 0.22.0	Yes (speaker_wav)	language=en	22 kHz

Chatterbox and Kokoro used Python 3.12 (venv-new). XTTS-v2 required Python 3.11 (venv-xtts) due to Coqui TTS dependency constraints. Random seed = 42 fixed across all models.

3. Feature Extraction

Three acoustic proxies for breathiness were extracted per WAV file using Praat (via parselmouth) and librosa:

Metric	Type	What it measures	Breathy direction
CPP (Cepstral Peak Prominence)	Primary	Periodicity loss in cepstral domain	Lower

HNR (Harmonic-to-Noise Ratio)	Supporting	Noise energy relative to harmonic energy	Lower
Spectral tilt	Supporting	High-frequency roll-off (100-5000 Hz)	Lower

Voiced frames are isolated via energy threshold + F0 pyin (librosa) with voiced probability ≥ 0.50 . Only voiced intervals ≥ 0.10 s contribute to metric means.

4. Gate Mechanism

Reference clips must pass three checks before analysis is allowed:

1. CPP separates breathy from neutral with Cohen's $d \geq 0.80$ and correct direction
2. At least one supporting metric separates with $d \geq 0.30$ and correct direction
3. All paired breathy-neutral contrasts are directionally consistent

5. Scoring

Score = mean_scaled_abs_delta - mean_contrast_retention. Lower score = better. Distance uses robust IQR-based scaling. Contrast retention clips to [0, 1] per metric.

Gate Results

Gate status: **PASSED**

All three pairs separated breathy from neutral with large effect sizes on both CPP and HNR.

Metric	Breathy mean	Neutral mean	Cohen's d	Direction	Paired OK	Pass?	Effect size
CPP (primary)	10.39	14.52	-4.15	✓	✓	PASS	
HNR	19.65	21.67	-2.03	✓	✓	PASS	
Spectral tilt	-7.38	-8.54	0.76	✗	✗	FAIL	

Thresholds: primary $d \geq 0.80$, supporting $d \geq 0.30$. CPP ($d=-4.15$) and HNR ($d=-2.03$) both pass. Spectral tilt fails direction check but is not required — only 1 of 2 supporting metrics needed.

Paired Contrasts (per pair, reference only)

Pair	Δ CPP (breathy - neutral)	Δ HNR (breathy - neutral)	Δ Tilt (breathy - neutral)
001 (p240 vs p229)	-2.85 ✓	-3.32 ✓	+1.36 ✗
002 (p253 vs p301)	-4.51 ✓	-1.63 ✓	+2.61 ✗
003 (p264 vs p282)	-5.03 ✓	-1.12 ✓	-0.47 ✓

All three CPP contrasts are negative (correct direction). All three HNR contrasts are negative. Spectral tilt is inconsistent across pairs — consistent with known limitations of tilt as a breathiness proxy.

Model Rankings

Lower scaled distance is better. Higher contrast retention is better. Lower score is better. Score = distance – retention.

Rank	Model	Outputs	Distance	Retention	Score	Score (lower = better)
1	XTTS-v2	6	0.615	0.518	0.097	
2	Chatterbox	6	0.822	0.510	0.311	
3	Kokoro (no cloning)	6	1.448	0.000	1.448	

Kokoro has no voice cloning — uses default voice (af_bella) for all generations. Zero contrast retention is expected. Included as baseline showing "no adaptation" behavior.

Per-Metric Distances

Mean scaled absolute delta per metric (lower = closer to reference):

Model	CPP		HNR		Tilt	
XTTS-v2	0.511		1.031		0.303	
Chatterbox	0.310		1.462		0.692	
Kokoro	0.778		2.664		0.903	

Chatterbox scores best on CPP distance (0.310). XTTS scores best on HNR (1.031) and spectral tilt (0.303). Kokoro consistently worst due to lack of voice adaptation.

Per-Pair Contrast Retention

How well each model preserves the breathy-neutral contrast per pair. Retention = $\min(1, \max(0, \text{ratio}))$.
Average across metrics within pair.

XTTS-v2

Pair	CPP ratio	HNR ratio	Tilt ratio	Mean retention	
001	0.171	-0.220	1.626	0.390	
002	0.439	0.726	0.687	0.617	
003	0.353	0.645	0.640	0.546	

Chatterbox

Pair	CPP ratio	HNR ratio	Tilt ratio	Mean retention	
001	0.146	0.162	1.475	0.436	
002	0.628	-0.044	0.771	0.466	
003	0.886	-1.201	1.161	0.629	

Kokoro

Pair	CPP ratio	HNR ratio	Tilt ratio	Mean retention	
001	0.000	0.000	0.000	0.000	
002	0.000	0.000	0.000	0.000	
003	0.000	0.000	0.000	0.000	

Chatterbox achieves best CPP retention on pair 003 (0.886). XTTS shows most consistent performance across all three pairs.
Kokoro zero retention is expected — see methodology.

Key Findings

Finding 1: XTTS-v2 preserves breathiness best overall

XTTS achieves the lowest score (0.097) driven by the lowest mean scaled distance (0.615) and highest mean contrast retention (0.518). Its performance is consistent across all three reference pairs, suggesting robust breathiness transfer regardless of accent or speaker identity.

Finding 2: Chatterbox competitive on retention, higher distance

Chatterbox matches XTTS on contrast retention (0.510 vs 0.518) but has 33% higher absolute distance from references (0.822 vs 0.615). This suggests Chatterbox preserves the direction of breathiness contrast but with less absolute accuracy — the output breathiness is in the right ballpark but farther from the original reference values.

Finding 3: Kokoro baseline confirms no cloning = no retention

Kokoro has no voice cloning capability. All outputs use the same default voice regardless of reference. As expected, contrast retention is zero and distance is worst (1.448). This validates the benchmark's sensitivity — it correctly detects when a model cannot adapt to reference breathiness.

Finding 4: Gate mechanism works correctly

The gate rejected the first two attempts (synthetic references, then weak VCTK pairs) and only passed after objective CPP+HNR ranking identified extreme outliers. The final gate values (CPP $d=-4.15$, HNR $d=-2.03$) provide strong confidence that the reference breathiness signal is real and measurable.

Finding 5: Spectral tilt is unreliable as a breathiness proxy

Across all three pairs, spectral tilt direction was inconsistent (correct on pair 003, wrong on 001 and 002). This supports existing literature suggesting tilt is sensitive to microphone placement, vocal tract length, and phonetic content rather than breathiness specifically. Future versions should consider replacing it with H1-H2 or harmonics-to-noise ratio from Praat.

Appendix: VCTK Speaker Ranking

All 62 female VCTK speakers were processed. 50 passed voiced_ratio > 0.15. CPP and HNR extracted on best sentence (001-010) by highest voiced ratio. Sorted by combined CPP+HNR z-score.

Only Southern England speakers initially selected (p240, p229, p225, p257, p228, p268). Expanded to full female pool when Southern England produced only 2 valid pairs. p264 (Scottish) selected as 3rd breathy candidate.

Final Pair Summary

Pair	Breathy	CPP	HNR	Modal	CPP	HNR	Sentence	CPP Δ	HNR Δ
001	p240 (S. England)	10.52	18.08	p229 (S. England)	13.36	21.41	s002	2.84	3.33
002	p253 (Welsh)	11.14	20.49	p301 (American)	15.65	22.12	s005	4.51	1.63
003	p264 (Scottish)	9.52	20.37	p282 (English NE)	14.55	21.48	s006	5.03	1.11

Output Files

File	Description
features/features.csv	24 rows, one per audio file with all metrics
results/gate_check.json	Gate decision and effect sizes per metric
results/per_sample_preservation.csv	Output-reference distances per sample
results/paired_contrast_preservation.csv	Contrast retention by model and pair
results/model_rankings.csv	Model-level summary sorted by score
results/report.md	Human-readable Markdown report
vctk_s002_ranked.csv	All speakers ranked by CPP for sentence 002
vctk_ranked_all.csv	All Southern England speakers ranked by CPP+HNR

Caveats

- CPP, HNR, and spectral tilt are acoustic proxies, not direct perception measures
 - Unmatched accent pairs (pair 002, 003) may introduce phonetic confounds
 - Minimum viable pipeline: 3 pairs, 3 models, 24 files — increase both for publication
 - Kokoro lack of voice cloning prevents fair comparison — include as baseline only
 - Single sentence per pair limits generalization to broader prosodic variation
 - Reference breathiness labels verified by perceptual listening check
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Dataset

VCTK Corpus 0.92: <https://datashare.ed.ac.uk/handle/10283/3443>